

SENTIVRA

One Security Layer. Every Threat.

Explainable, multi-layer threat detection that keeps your data where it runs.

Files

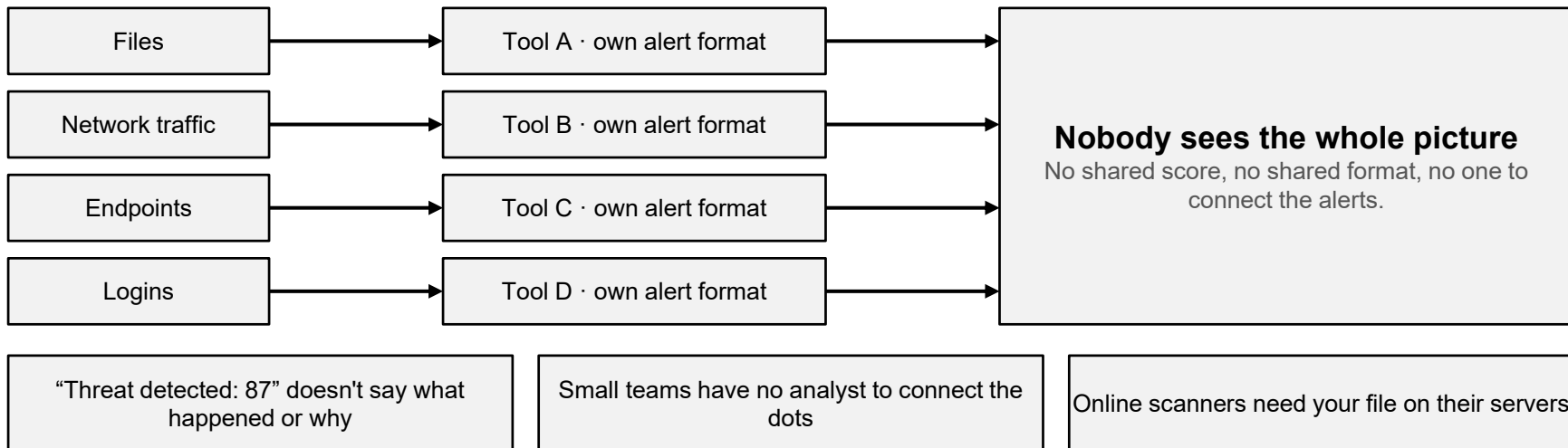
Network

Endpoints

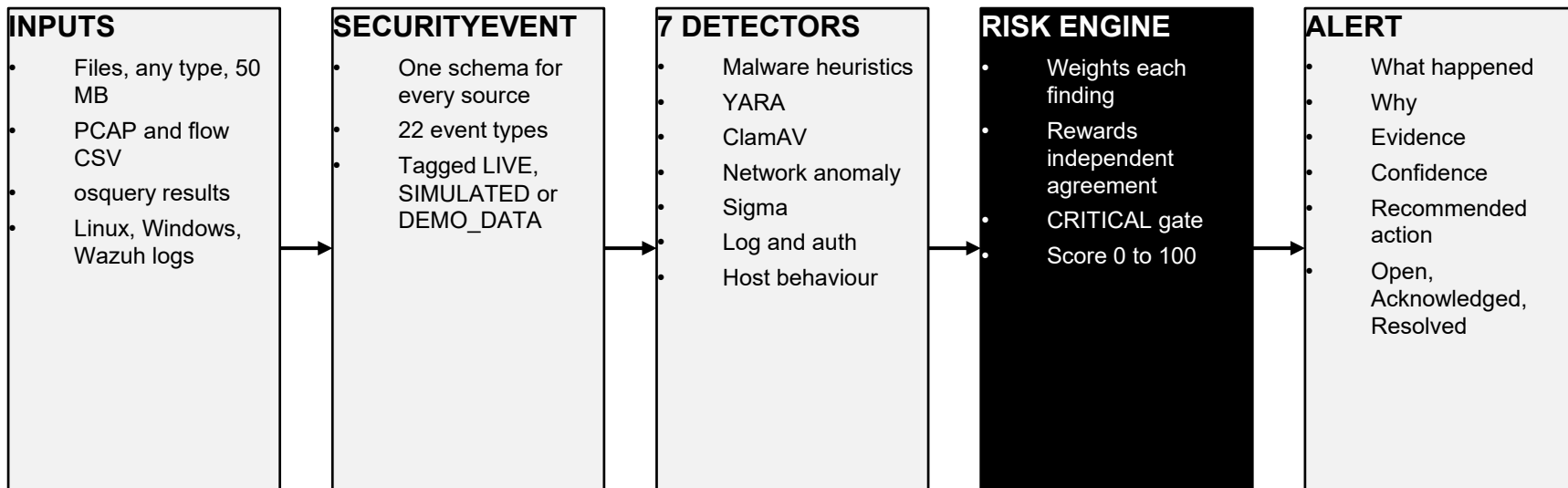
Logins

Attackers use every door. Defenders watch each one with a different tool.

HOW CAN an organisation with no security team get one explainable verdict across files, network traffic, endpoints and logins, without sending its data to anyone else?



One pipeline from raw data to an alert that explains itself



FastAPI · Next.js 16 dashboard, 14 pages · SQLite · append-only audit log. Detectors run concurrently, and a failing detector is logged and skipped. SENTIVRA detects and explains. A person decides.

What it catches today

Files

- Executable renamed .pdf
- Packed or encrypted (entropy 7.5+)
- Valid as two formats at once
- YARA signatures
- ClamAV, if installed

Never executed, never stored

Network

- Port scans, host sweeps
- Brute force and floods
- DNS tunneling, beaconing
- Bulk outbound transfer
- Four-layer ML scoring

PCAP or flow CSV

Endpoints

- Office app starts a shell
- Unsigned program calling out
- New persistence, rogue listener
- Ransomware-like file changes
- Security log cleared

osquery, 8 Sigma rules

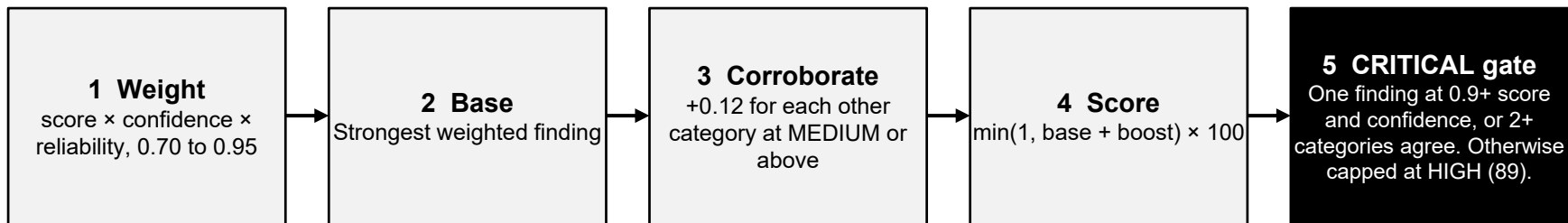
Logins

- Password guessing
- Password spraying
- Success after failures
- Wazuh alerts, level 7+

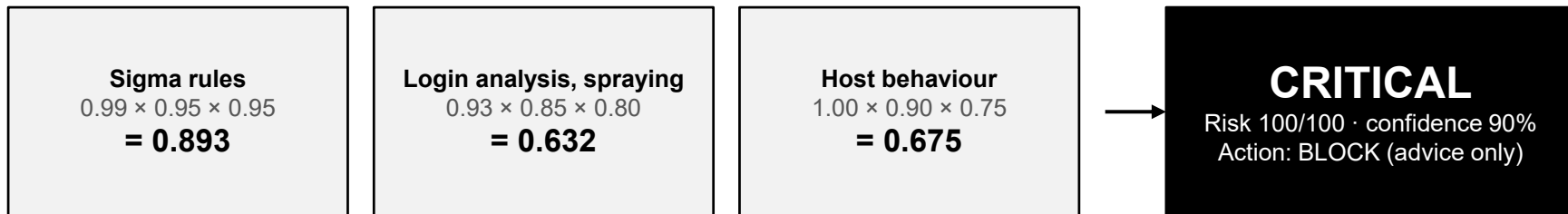
Linux, Windows, Wazuh logs

Built on open standards: YARA · Sigma · osquery · MITRE ATT&CK (858 techniques) · Wazuh

No single detector decides

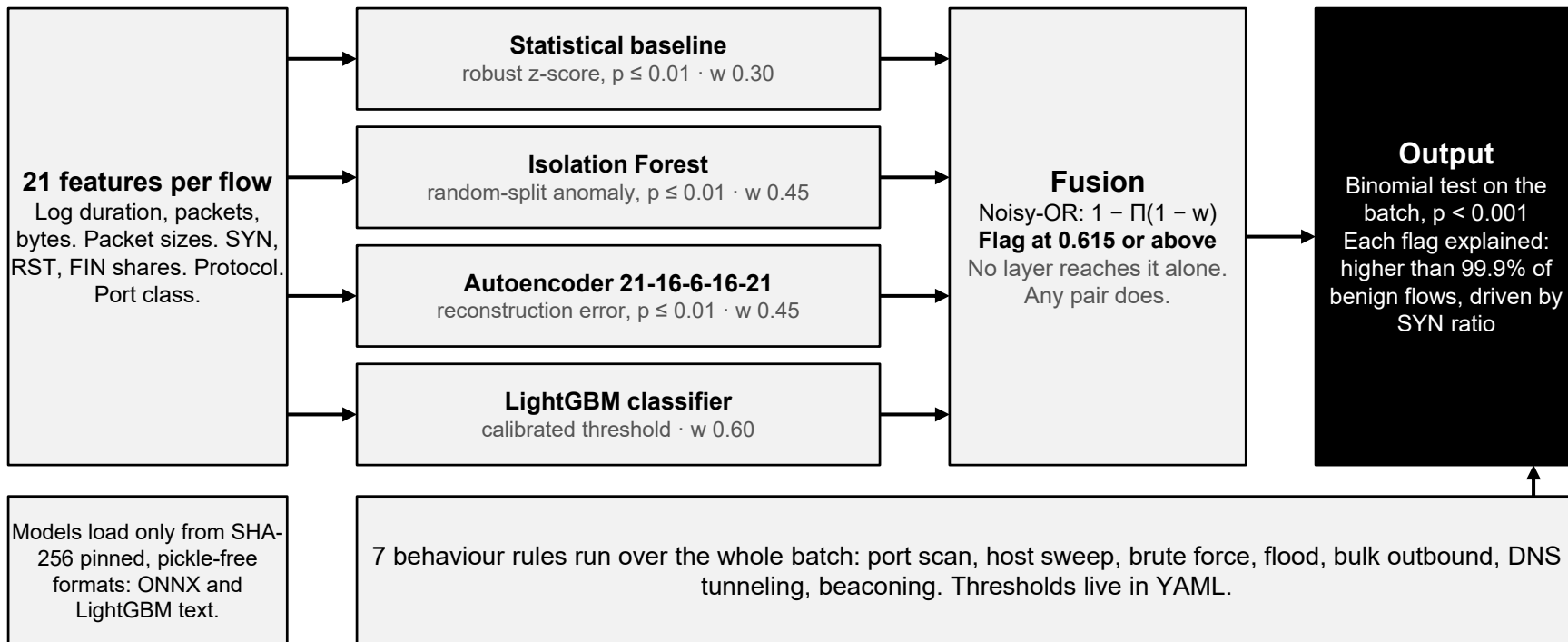


WORKED EXAMPLE · alert from the simulated fleet

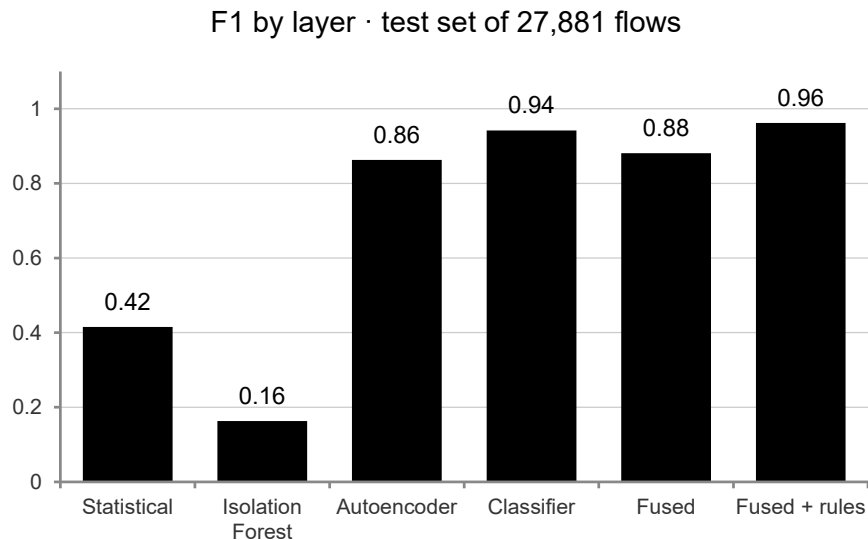


Base $0.893 + 2 \times 0.12$ corroboration = 1.133, capped at 1.0. Three categories agree, so the gate passes.

Four network models that must agree



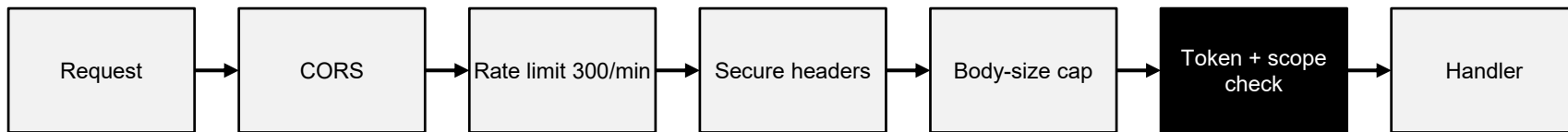
Measured, with the limits stated



Layer	Prec.	Recall	FPR
Statistical	0.704	0.294	1.10%
Isolation Forest	0.467	0.099	1.00%
Autoencoder	0.878	0.849	1.05%
Classifier	0.890	1.000	1.11%
Fused models	0.915	0.849	0.70%
Fused + rules	0.927	1.000	0.70%

All of this is synthetic and simulated data. It shows the pipeline, leakage controls and calibration work, not performance on real traffic. Endpoints: all 9 attack scenarios caught in 5 of 5 test fleets, and 0 findings on 10 benign fleets.

Built so the security tool isn't the weak point



Kept

- File hash, size, type, verdict
- Flagged-flow summaries as alert evidence
- Audit log: who did what, when, append-only

Discarded

- Uploaded file bytes
- Packet captures and flow files
- osquery rows and log lines
- Client secrets: only a hash is stored

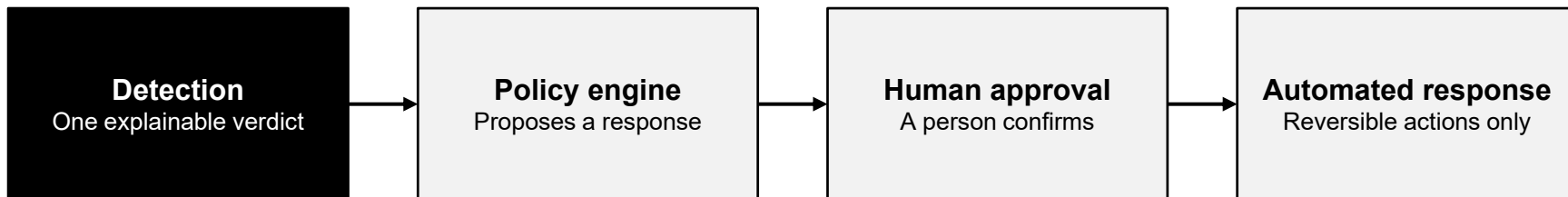
Self-defence

- Files parsed, never executed
- ZIP bomb and size limits
- Regex timeouts against ReDoS
- Models verified by SHA-256, no pickle
- 204 tests, clean bandit and pip-audit

From detection to protection



THE PATH TO AUTOMATED RESPONSE



Never on a single detector, never on demo data, with a kill switch that defaults to off.

Thank you

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Live demo: Demo mode, then Run all scenarios.